

System Log Forensic Analysis using Python

Python Program:

```
import re
from collections import Counter

with open("login_attempts.log", "w") as f:
    f.write("2026-07-08 09:00:01 LOGIN SUCCESS user=alice ip=10.0.0.5\n")
    f.write("2026-07-08 09:01:15 LOGIN FAILED user=admin\n")
    f.write("2026-07-08 09:01:20 LOGIN FAILED user=admin\n")
    f.write("2026-07-08 09:01:25 LOGIN FAILED user=admin\n")
    f.write("2026-07-08 09:01:30 LOGIN FAILED user=admin\n")
    f.write("2026-07-08 09:01:35 LOGIN FAILED user=admin\n")
    f.write("2026-07-08 09:02:00 LOGIN SUCCESS user=bob ip=10.0.0.8\n")

failed_by_ip = Counter()
with open("login_attempts.log") as f:
    for line in f:
        if "LOGIN FAILED" in line:
            match = re.search(r"ip=(\S+)", line)
            if match:
                failed_by_ip[match.group(1)] += 1

print("System log forensic analysis - failed logins by source IP:")
for ip, count in failed_by_ip.items():
    print(f"{ip}: {count} failed attempts")
    if count >= 5:
        print(f"Evidence of brute-force intrusion attempt from {ip}")
```

Sample Output:

```
System log forensic analysis - failed logins by source IP:
203.0.113.99: 5 failed attempts
Evidence of brute-force intrusion attempt from 203.0.113.99
```

Explanation:

This program performs a simple forensic analysis of a system login log. It counts failed login attempts from each IP address using regular expressions and the Counter class. If an IP address has five or more failed login attempts, the program flags it as a possible brute-force intrusion attempt. This demonstrates a basic log analysis technique used in digital forensics and cybersecurity.